

ASHMIT DEB

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EDUCATION

University of Michigan

Aug. 2023 – Dec. 2027

B.S. in Computer Science

Ann Arbor, MI

Cumulative GPA: 3.4/4.0

Leadership: Michigan Financial and Math Society (Quant Lead) & Michigan Student AI Laboratory (AI Project Director)

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Computer Organization, Data Science and Statistical Analysis, User Interface Development, Computer Vision, Software Engineering

EXPERIENCE

Software Engineer Intern

May 2026 – Present

Health Care Service Corporation

Chicago, IL

- Developed **Python** and **PySpark** data pipelines for **ML**-driven infrastructure capacity planning, analyzing CPU, RAM, storage, and membership utilization trends to support mainframe, **OS**, and distributed-system upgrade decisions.
- Built multi-model forecasting and imputation workflows over irregular historical telemetry, improving reliability of capacity signals and **AI**Ops reporting across **Linux**, server, application, and enterprise infrastructure portfolios.
- Automated problem reporting and infrastructure inventory analysis across **2,700+** enterprise applications using **SQL**, **ServiceNow** data, and operational dashboards to improve incident visibility and reduce manual triage.

Software Engineer Intern

May 2025 – Aug. 2025

Sigray

Benicia, CA

- Enhanced development processes in **Python** with **C#** spectral-analysis modules and **NUnit** tests for a microscopic X-Ray boosting feature while enabling **CI/CD** validation via **Jenkins** on **Azure AKS** (Remote Access API + **GUI**).
- Integrated Machine Learning spectral decomposition by implementing **k-means clustering** in Jupyter, thereby embedding fully automated mineral-phase segmentation directly into the scalable data analysis pipeline.
- Built a **REST Node.js** microservice on **AWS EC2** to stream real-time scanned X-Ray data to a **Vue.js** frontend dashboard, cutting preview latency by **30%** via telemetry tuning and doubling productivity among engineers.

Software Engineer Intern

May 2024 – Aug. 2024

Blue Cross Blue Shield of Michigan

Detroit, MI

- Designed a **REST** claims processing system with **Java** and **Spring Boot** integrating **PostgreSQL** validation logic to power an **Angular** frontend, reducing processing times by **40%** through scalable workflows.
- Designed and tuned **MySQL** schemas in **AWS RDS**, adding unit-tested queries and **SharePoint** validation dashboards, boosting data integrity and automated schema migrations for repeatable zero downtime releases.
- Expanded the iOS and Android membership app by improving the eligibility/benefits-lookup feature through fixing **15+ bugs**, integrating the **React Native** UI through **Firebase** to a **Spring Boot** backend for further security.

PROJECTS

Reddit API Stock Analyzer

Jan. 2025 – Feb. 2025

- Developed a multithreaded **C++** app that pulls finance-subreddit posts (r/WSB, r/stocks, r/investing) via the Reddit API (**cURL** + **JsonCpp**), scores sentiment, and outputs to a live **CLI** and auto-generated **HTML** report in real time.
- Applied modern **C++** practices (**RAII**, **STL**) and thread-pool concurrency with mutex-guarded queues, plus lock-free metrics, to accelerate data ingestion and scoring, producing near-real-time, actionable stock-sentiment insights.

Real-Time Earthquake Monitor

Sep. 2024 – Nov. 2024

- Built and operated a **Node.js** and **Express.js** ingestion service with **Socket.io**, designing resilient real-time data pipelines while working in an **Agile** and **Scrum** SDLC with code reviews, issue tracking, and regular releases.
- Used **HTML**, **CSS**, **JavaScript**, and **React** on frontend and deployed **PostgreSQL**, Node API, and **Flask** analytics with **Docker** Compose pushing images to **AWS ECR** and securely scaling on **ECS** for continuous seismic tracking.

SKILLS

Languages: C#, C++, C, Swift, Python, Java, SQL, JavaScript, TypeScript, Go, HTML, CSS, Kotlin, Rust, Perl

Frameworks/Libraries: Node.js, Vue.js, React.js, Angular, .NET, Django, TensorFlow, PySpark, MongoDB

Cloud/Tools: Azure AKS, AWS RDS/EC2/ECS, NGINX, Git, MySQL, Docker, Kubernetes, Jenkins, JIRA, Linux

Interests: NFL analytics, Competitive Tennis, Golf, Traveling, Cultural Exploration, Photography, Bengali